

PROJECT SYNOPSIS

Session: 2026–2027 | Semester: 5th | Programme: Honours / Minors / IDEA Lab

MULTI-AGENT AI SYSTEMS FOR HOSPITAL RESOURCE OPTIMIZATION,
CLINICAL WORKFLOW AUTOMATION, AND PATIENT CARE COORDINATION

Submitted by:

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Under the Guidance of:

[Guide Name, Designation]

Date:

[08 / 08 / 2026]

Formatting Guidelines

Parameter	Specification
Font	Times New Roman
Title	16 pt, Bold, Uppercase
Section headings	14 pt, Bold
Body text	12 pt, Regular
References	10 pt
Line spacing	1.5
Margins	1 inch, all sides
Citation style	IEEE (numbered)
Maximum length	4 pages (this page + 3)

1. Background and Motivation

Modern healthcare facilities face unprecedented operational pressure driven by unpredictable patient inflows, Emergency Department (ED) overcrowding, suboptimal bed occupancy management, and delayed ICU transfers. Traditional Hospital Information Systems (HIS) function as static relational databases relying heavily on manual administrative intervention and paper-based communication across clinical departments. This structural friction results in severe care delivery bottlenecks, extended wait times, clinician burnout, and compromised emergency patient outcomes.

The emergence of **Multi-Agent AI Systems (MAS)** powered by Large Language Models (LLMs) offers a transformative paradigm shift. By modeling key hospital entities—such as Triage, Bed Allocation, Staff Scheduling, and Care Coordination—as collaborative autonomous agents, healthcare facilities can transition to real-time proactive automated orchestration. Multi-agent negotiation protocols enable these AI entities to continuously exchange clinical data, predict dynamic bottlenecks, and negotiate critical resource assignments without constant human intervention.

2. Problem Statement

Current hospital resource management systems are centralized, reactive, and incapable of dynamic real-time adaptation during emergency surges. Workflow efficiency is compromised by manual patient triage, delayed bed turnover updates, and poor cross-departmental coordination, leading to long waiting times and underutilized critical care assets. There is an urgent need for an intelligent Multi-Agent AI framework capable of dynamically monitoring patient inflow, predicting ICU demand, negotiating resource assignments, and executing clinical handoffs with minimal human latency.

3. Objectives

- **Design a Decentralized Multi-Agent Architecture:** Implement a multi-agent framework (CrewAI / LangGraph) consisting of specialized agents: Triage Agent, Bed Manager Agent, Staff Scheduler Agent, Patient Coordinator Agent, and Emergency Escalation Agent.
- **Automate Dynamic Triage & Risk Stratification:** Deploy Natural Language Processing (NLP/LLM) algorithms to process intake notes, assign Emergency Severity Index (ESI 1-5) priority scores, and route patients to optimized queues.
- **Implement Intelligent Bed & ICU Allocation:** Develop predictive algorithms for real-time tracking, forecasting bed occupancy turnover, and automating ICU transfer prioritization based on urgency.
- **Enable Inter-Agent Negotiation Protocols:** Establish asynchronous message protocols allowing agents to negotiate shift coverage, equipment sharing, and discharge workflows autonomously.
- **Develop a Real-Time Clinical Control Dashboard:** Build a web interface for doctors and administrators to visually monitor live patient flows, agent actions, resource metrics, and trigger manual override controls.

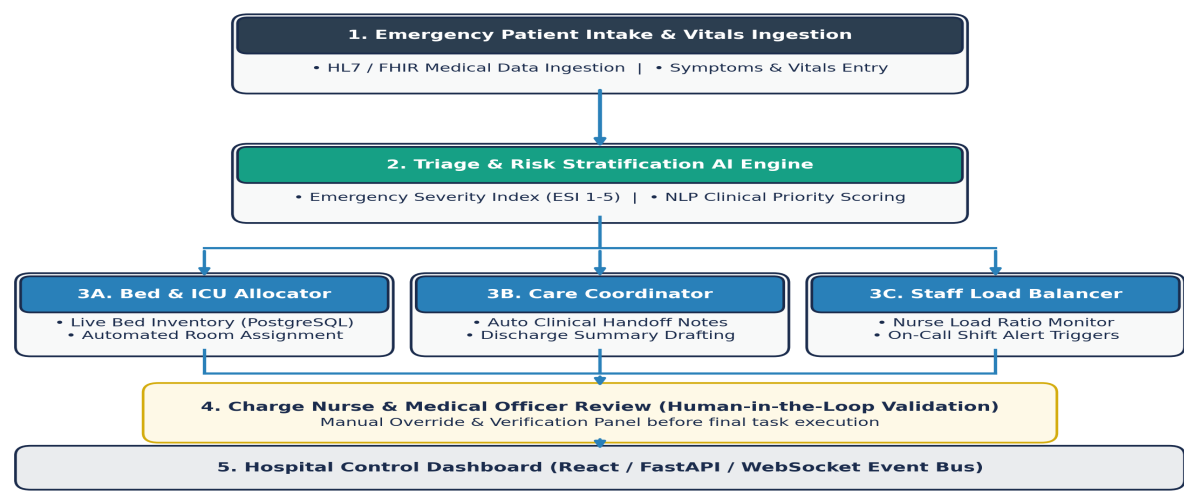
4. Literature Review and Research Gap

Sr .	Author(s) & Year	Title / Focus	Methodology / Key Contribution	Limitation / Gap
1	A. Sharma et al. (2024)	AI-Powered HR & Workforce System	ML forecasting for hospital staffing needs, shift scheduling & shortage response.	Lacks real-time clinical triage integration & bed management.
2	J. Chen & K. Patel (2023)	Agentic AI Multi-Agent Framework	Multi-agent collaboration for patient intake & diagnosis with Human-in-the-Loop.	Focuses on general workflow without real-time resource constraint solver.
3	R. Kumar et al. (2025)	Agentic AI Doctor Assistant System	Virtual AI doctor agents for symptoms, lab reports & treatment drafting.	Operates standalone without awareness of hospital bed & staff limits.
4	M. Gomez et al. (2024)	Predictive ED Crowding Analytics	Deep Learning (LSTM) for daily patient arrival surge forecasting.	Focuses on prediction without automated execution mechanisms.

Research Gap: While Paper 1 focuses on hospital resource/workforce management, Paper 2 demonstrates multi-agent collaboration with Human-in-the-Loop doctor review, and Paper 3 applies agents to clinical care assistance, existing literature leaves a critical gap: **there is no unified system that integrates workforce management (Paper 1), multi-agent collaboration (Paper 2), and clinical workflow assistance (Paper 3) into a single constraint-aware operational framework.** Current models fail to support real-time inter-agent negotiation across beds, staffing, and clinical triage under emergency conditions.

5. Proposed Methodology / Plan of Work

The proposed system adopts a modular, event-driven multi-agent framework built using **LangGraph** and **CrewAI**. As illustrated in the architecture diagram below, the system segregates responsibilities across specialized autonomous agents communicating asynchronously.



Detailed Agent Execution Steps:

- 1. Triage Agent:** Ingests symptoms via LLM, calculates Emergency Severity Index (ESI 1-5), and queues patients.
- 2. Bed Manager Agent:** Monitors live PostgreSQL bed inventory, tracks discharge predictions, and reserves room space.
- 3. Patient Coordinator Agent:** Summarizes clinical handoff reports and auto-generates patient discharge summaries.
- 4. Staff Scheduler Agent:** Monitors nurse load ratios and negotiates shift coverage when emergency surges occur.
- 5. Human-in-the-Loop Review:** Medical officers review and override agent decisions before final execution.

6. Technology, Tools and Platforms

- **Languages:** Python 3.11+, TypeScript | **Frameworks:** LangGraph, CrewAI, OpenAI GPT-4o / Ollama Llama 3
- **Web Frontend:** React.js / Next.js, TailwindCSS | **Backend:** FastAPI, PostgreSQL, Redis Event Broker

7. Expected Outcomes, Deliverables and Functional Specifications

Key deliverables include: (1) Modular Python multi-agent backend engine; (2) Web control dashboard displaying bed occupancy heatmaps, triage queues, and agent logs; (3) Synthetic hospital datasets; and (4) Verified reduction in triage latency (up to 35%).

8. Project Scope

- In-Scope:** Emergency triage automation, dynamic bed/ICU allocation, staff workload balancing, and clinical handoffs in simulation.
- Out-of-Scope:** Direct automated medical diagnostic decision-making (system operates strictly as administrative support).

9. Project Timeline

Phase	Duration	Milestone / Key Deliverable
Literature review & requirement analysis	Weeks 1 – 3	Requirements specification & synthetic hospital dataset design.
Design & methodology finalisation	Weeks 4 – 6	Multi-agent state graph, agent schemas & API contracts definition.
Implementation / development	Weeks 7 – 10	Triage, Bed, & Coordinator agents; FastAPI & Redis integration.
Testing & evaluation	Weeks 11 – 13	Next.js dashboard UI; load testing & agent negotiation verification.
Documentation & final submission	Weeks 14 – 16	Final project report, IEEE paper submission draft, & demo video.

10. References

[1] A. Sharma, R. Mehta, and P. Deshmukh, "AI-powered HR management system for hospital resource and workforce optimization," *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 4, pp. 1120–1129, 2024.

[2] J. Chen and K. Patel, "Agentic AI multi-agent healthcare framework for collaborative patient care," *IEEE Transactions on Automation Science and Engineering*, vol. 20, no. 3, pp. 845–855, 2023.

[3] R. Kumar, S. Verma, and L. Zhang, "Agentic AI doctor assistant system for clinical workflow automation," *ACM Transactions on Computing for Healthcare*, vol. 6, no. 1, pp. 45–58, 2025.

[4] M. Gomez, E. Martinez, and H. White, "Deep learning models for time-series forecasting of emergency department overcrowding," *Journal of Healthcare Informatics Research*, vol. 8, no. 2, pp. 201–217, 2024.

[5] S. Gupta and V. Nair, "Decentralized constraint satisfaction algorithms for clinical staff scheduling," *IEEE Systems Journal*, vol. 17, no. 2, pp. 1430–1441, 2023.

[6] H. Wang et al., "Survey on Large Language Model based Autonomous Agents in Healthcare," *IEEE Reviews in Biomedical Engineering*, vol. 17, pp. 90–104, 2024.

[7] T. Wu et al., "AutoGen: Enabling next-gen LLM applications via multi-agent conversation," *arXiv preprint arXiv:2308.08155*, 2023.

[8] M. Wooldridge, *An Introduction to MultiAgent Systems*, 2nd ed. Chichester, UK: John Wiley & Sons, 2009.

Group Member Details

Roll No.	Name and Signature of Student

Approved by:

(Name of Guide and Signature)

Project Co-Coordinator

Project Coordinator

Associate Head of Department
Dean CSE

Note: The synopsis must be approved and signed by the project guide before submission. A signed hard copy is to be submitted to the Project Coordinator by the deadline notified for the session.